Symptom Based Dynamic Disease Spreading : Agent Base Model Approach

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Abstract

This research introduces a novel, complex, stochastic, and symptom-based epidemiological model designed to overcome limitations of traditional compartmental models. Unlike conventional approaches that rely on fixed parameters and homogeneous populations, our model dynamically simulates disease progression through a complex network of interacting symptoms, where severity is continuously recalculated based on inter-symptom relationships. Infectiousness and recovery rates are not static but evolve with individual symptom states and synergistic interactions. The model incorporates agent heterogeneity through varying immunity levels, health reserves, and probabilistic outcomes for recovery and death, offering a more biologically realistic representation of disease spread. This framework provides a next-generation simulation platform grounded in complex adaptive systems and network science, enhancing our understanding of personalized disease trajectories and the integration of social infrastructure in epidemiological dynamics.

1 Introduction

Epidemic modeling has long been a cornerstone of public health, providing crucial insights into disease transmission dynamics and informing intervention strategies. From the foundational deterministic compartmental models to more recent sophisticated approaches, the evolution of these models reflects a continuous effort to capture the intricate complexities of real-world epidemiological phenomena. Early efforts, such as the seminal Susceptible-Infectious-Recovered (SIR) model, laid the groundwork for understanding population-level disease dynamics [KM27]. The SIR model, and its extensions like the Susceptible-Exposed-Infectious-Recovered (SEIR) model, simplify populations into homogeneous compartments, assuming uniform mixing and fixed transition rates between states [AM91]. While these models have been instrumental in providing a macroscopic view of epidemics, their inherent assumptions of homogeneity and determinism often fall short in capturing the nuanced realities of disease spread, particularly in diverse and interconnected populations.

The limitations of classical compartmental models spurred the development of more granular approaches, notably agent-based models (ABMs) and network-driven models. ABMs simulate individual agents and their interactions, allowing for the incorporation of heterogeneity in demographics, behaviors, and spatial locations [Eps09]. This individual-level granularity enables a more realistic representation of disease transmission pathways and the impact of localized interventions. Concurrently, the application of complex network theory to epidemiology has revolutionized our understanding of how contact patterns influence disease propagation [PSV01]. By representing individuals as nodes and their interactions as edges in a network, these models can explore the effects of heterogeneous contact structures, super-spreaders, and community dynamics on epidemic trajectories. Tools like NetworkX [HSS08] have become indispensable in constructing, manipulating, and analyzing these complex networks, facilitating the study of disease spread on realistic social and spatial topologies.

Despite these advancements, a significant gap remains in models that fully integrate the dynamic and personalized nature of disease progression at the individual level, particularly concerning symptomology and physiological responses. Traditional models often treat infected individuals as a monolithic group with uniform infectiousness and recovery profiles. However, clinical reality demonstrates that disease manifestation is highly variable, influenced by individual physiological states, immune responses, and the complex interplay of symptoms. This variability is crucial for understanding disease severity, transmission potential, and the effectiveness of personalized medical interventions. The need for

models that bridge the gap between population-level dynamics and individual-level biological realism is increasingly evident, especially in the context of emerging infectious diseases with diverse clinical presentations.

This research presents a novel epidemiological model that fundamentally diverges from traditional approaches by embracing complexity, stochasticity, and a symptom-based logic. Unlike conventional SIR or SEIR models, which categorize populations into homogeneous compartments and rely on fixed parameters for infection (β) and recovery (γ) rates, our model adopts an individual-based, agent-level simulation. Each agent in our model experiences a personalized disease progression, shaped by their unique dynamic symptom profile, immunity class, and health reserve. This approach moves beyond the simplistic assumption of uniform infectiousness among all infected individuals, allowing for a more nuanced representation of disease transmission based on individual clinical states.

Our model treats various symptoms not as isolated indicators but as a complex network structure where they interact with one another. The severity of each symptom is continuously recalculated based on its relationship with other symptoms, reflecting the synergistic and antagonistic interactions observed in physiological systems. Symptoms are categorized according to their associated physiological systems, and both intra-system and inter-system interactions are weighted differently, allowing for a detailed representation of multi-systemic disease effects. Crucially, infectiousness and recovery rates are not fixed parameters; instead, they are dynamically updated based on the current symptom states, symptom severity, and synergistic interactions between symptoms. This dynamic parameterization provides a significant departure from the static nature of traditional models, enabling a more adaptive and realistic simulation of disease progression.

Furthermore, the model incorporates a high degree of individual heterogeneity and stochasticity. The population is classified according to different immunity levels, and each individual is randomly assigned a unique health reserve, recovery thresholds, and starvation times. This probabilistic assignment makes recovery and death outcomes stochastic processes, reflecting the inherent randomness and variability in biological systems. Consequently, the spread and progression of the disease are determined not merely by a simple infection-recovery cycle but by the dynamic interaction of multiple factors, including the complex network of symptom interactions, the randomness in individual parameters, and the capacity of the healthcare system. This integration of individual-level stochasticity and dynamic parameters allows our model to exhibit complex adaptive behavior, driven by daily-varying β and γ values and multiple feedback loops, moving beyond the deterministic outcomes of conventional models.

The most significant difference lies in the model's ability to simulate the temporal evolution of disease through symptom networks and physiological system interactions. It performs synergy computations between symptoms using complex network analysis, leveraging the capabilities of NetworkX to capture intricate relationships. The model captures individual variation through immune system diversity and the concept of health reserves, providing a more biologically plausible representation of individual resilience and susceptibility. Moreover, it integrates social infrastructure constraints, such as hospital capacity and cafeteria interactions, which are often overlooked in simpler models. The inclusion of gradual health deterioration and dynamic recovery bonuses further enhances the realism of disease progression, allowing for a more accurate assessment of intervention strategies that target specific aspects of disease management.

This model is not merely a conventional epidemiological model but a next-generation simulation platform grounded in complex adaptive systems and network science principles. It enhances biological realism by incorporating detailed physiological and symptomatic dynamics, integrates social factors by considering healthcare infrastructure and individual behaviors, and supports personalized medicine approaches by simulating individual-specific disease trajectories. By situating our model within the broader fields of complex adaptive systems, personalized medicine, network epidemiology, and systems biology, we aim to provide a framework that can address contemporary challenges in public health with unprecedented detail and accuracy.

This research aims to introduce the theoretical underpinnings and methodological framework of this novel epidemiological model. Specifically, we detail the construction of the symptom network, the algorithms for dynamic parameter updates, and the mechanisms for incorporating individual heterogeneity and stochasticity. This model accomplishes what others cannot by providing a granular, dynamic, and biologically realistic simulation of disease progression that accounts for the complex interplay of individual physiological responses and environmental factors. Its methodological novelties

include the symptom-based network approach, dynamic interaction weights, and the integration of health reserves and probabilistic outcomes. Our primary assumption is that disease progression can be effectively modeled as a complex adaptive system driven by dynamic symptomatic interactions, offering a powerful tool for understanding and predicting epidemic trajectories in heterogeneous populations.

2 Methodology

2.1 Symptoms

In this model, there are 16 distinct symptoms that are likely to emerge during infectious diseases, and these symptoms have been categorized into five different groups based on their characteristics and potential impact on disease transmission dynamics.

Symptom	**B+**	**B-**	**B0**	**Zeta**	**System**	**Spreading Mechanism**
Sneezing	0.84	Undefine	Undefine	0.2	Respiratory	Droplet, Airborne
Cough	0.68	Undefine	Undefine	0.2	Respiratory	Droplet, Airborne
Runny Nose	0.41	Undefine	Undefine	0.2	Respiratory	Contact, Fomite
Diarrhea	0.23	Undefine	Undefine	0.3	Excretory	Fecal-Oral
Sepsis	Undefine	Kills Spreading Potential	Undefine	0.76	Immune	Restrictive
High Fever	Undefine	Kills Spreading Potential	Undefine	0.4	Immune	Restrictive
Loss of Consciousness	Undefine	Kills Spreading Potential	Undefine	0.79	Nervous	Restrictive
Coma	Undefine	Kills Spreading Potential	Undefine	0.85	Nervous	Restrictive
Pneumonia	Undefine	Kills Spreading Potential	Undefine	0.66	Respiratory	Restrictive
Fever	Undefine	Undefine	0	0.3	Immune	Undefine
Muscle Pain	Undefine	Undefine	0	0.35	Muscle	Undefine
Nausea	Undefine	Undefine	0	0.38	Digestive	Undefine
Vomiting	0.25	Undefine	Undefine	0.45	Digestive	Fomite
Sweating	Undefine	Undefine	0	0.32	Immune	Undefine
Skin Lesions	0.25	Undefine	Undefine	0.38	Skin/Venereal	Fomite

Table 1: Symptom classification and spreading characteristics

2.1.1 Classification Based on B Value and Transmission Mechanism

The transmission mechanism classification functions as the primary framework for symptom-based epidemiological modeling. Symptoms are grouped into four distinct transmission pathways: Airborne/Respiratory (e.g., sneezing $\beta = 0.84$, coughing $\beta = 0.68$, nasal discharge $\beta = 0.41$), Gastrointestinal (e.g., vomiting $\beta = 0.25$, diarrhea $\beta = 0.23$), Contact/Dermal (e.g., skin lesions $\beta = 0.25$), and Systemic/Inhibitory (e.g., fever, sepsis, pneumonia; classified with β^- due to their inhibitory nature). The beta coefficient (β) quantifies each symptom's contribution to disease transmissibility, with airborne symptoms assigned the highest values due to their capacity for active viral dispersal. In contrast, systemic symptoms are marked as β^-, indicating that they reduce transmission through behavioral isolation and decreased social activity. This classification feeds directly into the transmission rate (β) calculation within the SIR model: the presence of β^- symptoms transitions the model from high-transmission scenarios ($\beta_{combined} > 2.0$) to low-transmission states ($\beta \approx 0.001$–0.005), reflecting the epidemiological reality that severe illness often leads to decreased spread, despite potentially higher individual viral loads.

2.1.2 Severity Level Classification

The severity level classification employs zeta (ζ) coefficients to quantify symptom intensity and its influence on disease progression and recovery dynamics. Symptoms are categorized into three severity tiers:

Mild ($\zeta = 0.2$–0.35): Including common symptoms such as sneezing, cough, and fever that generally allow individuals to maintain normal social behavior. Moderate ($\zeta = 0.38$–0.45): Encompassing symptoms like high fever, vomiting, and nausea that cause noticeable physiological discomfort. Severe ($\zeta = 0.66$–0.85): Comprising life-threatening conditions such as pneumonia, sepsis, and loss of

consciousness. ζ values directly affect the calculation of γ (the recovery rate) through an inverse relationship—higher severity levels correspond to lower recovery rates and longer infectious durations. Additionally, the model incorporates synergistic severity effects by computing ζ_{net}, which is derived from physiological system weights and network connectivity. This enables the model to capture the dynamic interactions between symptoms, particularly when severe symptoms co-occur and compound overall disease burden.

This severity-based stratification provides essential input for healthcare resource allocation modeling and mortality risk assessment within the epidemiological simulation framework.

2.1.3 Physiological System Classification

The physiological system classification divides symptoms into seven distinct organ system categories to model inter-system interactions and patterns of disease progression. These systems include: Respiratory (sneezing, cough, pneumonia), Immune (fever, sepsis), Nervous (loss of consciousness, coma), Digestive (nausea, vomiting), Excretory (diarrhea), Muscular (muscle pain), and Skin/Venereal (skin lesions). Each system is assigned weighted connectivity coefficients that reflect biological dependencies.

The approach utilizes a physiological weight matrix (physio weights) where intra-system connections are assigned the maximum weight (e.g., 1.0 for respiratory–respiratory), while inter-system weights vary based on pathophysiological relevance (e.g., respiratory–nervous: 0.8, respiratory–immune: 0.6, immune–nervous: 0.6).

Symptoms affecting the same physiological pathway tend to exhibit synergistic effects in both severity (ζ_{net}) and transmissibility ($\beta_{combined}$) calculations.

2.1.4 β Calculation

The beta calculation methodology transforms qualitative symptom presentations into quantitative transmissibility metrics through a multi-stage computational framework. This approach integrates network topology analysis, physiological system interactions, and epidemiological scaling functions to produce biologically plausible transmission parameters for SIR model integration.

The calculation employs three distinct beta categories with fundamentally different computational pathways. Beta-enhancing symptoms ($\beta_{raw} > 0$) contribute positively to transmission through active viral dispersal mechanisms, receiving raw beta values ranging from 0.23 to 0.84 based on their transmission efficiency. Beta-inhibiting symptoms ($\beta^- = True$) override baseline transmission calculations by enforcing social isolation behaviors, reflecting severe illness states that reduce community contact regardless of viral load. Beta-neutral symptoms ($\beta_{raw} = 0$) serve as background clinical markers without direct transmission impact, contributing to disease severity without altering infectivity parameters.

The methodology employs four distinct scaling functions to transform individual symptom parameters into population-level epidemiological metrics:

1. **Sigmoid Scaling:**

$$\beta_{\text{SIR}} = \frac{\beta_{\text{combined}}}{1 + \beta_{\text{combined}}}$$

   This function bounds unbounded β values to the interval $[0, 1]$, making them suitable for integration into SIR models.

2. **Network Scaling:**

$$\beta_{\text{scaled}} = \beta_{\text{raw}} \times (\text{degree_centrality} + \alpha \times \text{system_proportion})$$

   This scaling amplifies the effect of individual symptoms based on their connectivity in the symptom network and the clustering within physiological systems.

3. **Synergy Scaling:**

$$\beta_{\text{combined}} = \sum \beta_{\text{scaled}} \times \left(1 + \theta \times \frac{n-1}{n}\right)$$

   This captures the super-additive (synergistic) effects when multiple symptoms co-occur.

4. **Severity Scaling:** Severity values (ζ) are transformed using a sigmoid function to dynamically adjust symptom intensity based on disease progression:

$$\zeta_{\text{scaled}} = \frac{\zeta}{1+\zeta}$$

When beta-minus symptoms are present in any symptom combination, the model enforces a transmission override mechanism where $\beta \in [0.001,\ 0.005]$ regardless of concurrent symptoms. This reflects the epidemiological reality that severe illness-induced behaviors (hospitalization, bed rest, social isolation) dramatically reduce transmission potential despite potentially high viral loads. The override mechanism captures the behavioral epidemiology component where disease severity inversely correlates with community transmission opportunity.

Network-Weighted Scaling in Beta-Enhancing Scenarios

For β enhancing scenarios, the methodology employs network-weighted scaling, defined as:

$$\beta_{\text{scaled}} = \beta_{\text{raw}} \times \left(\frac{\text{degree_centrality}}{\text{max_degree}} + \alpha \times \text{system_proportion} \right)$$

In this equation:

- degree_centrality quantifies the connectivity of a symptom within the overall symptom network.
- system_proportion represents the fraction of co-occurring symptoms that belong to the same physiological system.
- The scaling factor $\alpha = 0.4$ balances the influence of network topology with intra-system clustering effects.

Synergistic Combination of Scaled Beta Values

Individual scaled beta values undergo synergistic combination using the following equation:

$$\beta_{\text{combined}} = \sum \beta_{\text{scaled}} \times \left(1 + \theta \times \frac{n-1}{n}\right)$$

where:

- $\theta = 0.3$ represents the synergy coefficient,
- n denotes the number of active β enhancing symptoms.

This multiplicative enhancement captures the epidemiological observation that multiple concurrent symptoms often produce super-additive transmission effects, particularly when symptoms affect complementary transmission pathways (e.g., respiratory + gastrointestinal symptoms).

Final Epidemiological Parameter: β_{SIR}

The final epidemiological parameter β_{SIR} applies a sigmoid transformation to ensure bounded transmission rates suitable for integration into the SIR model:

$$\beta_{\text{SIR}} = \frac{\beta_{\text{combined}}}{1+\beta_{\text{combined}}}$$

This transformation prevents unrealistic transmission scenarios while maintaining biological plausibility across diverse symptom combinations and disease severity spectra.

The bounded output ensures numerical stability in differential equation solving, while preserving the relative ranking of transmission risk across different symptom profiles.

2.1.5 Calculation of γ

Gamma Calculation and Recovery Rate Modeling

The Gamma (γ) calculation employs a dynamic, ζ-dependent sigmoid transformation that converts symptom severity metrics into recovery rates suitable for SIR epidemiological modeling:

$$\gamma_{\text{SIR}} = \frac{1}{1+\zeta}$$

This methodology incorporates:

- Adaptive threshold mechanisms to account for varying disease severity,
- Multi-stage mathematical transformations that ensure non-linear, yet biologically plausible recovery dynamics.

The resulting γ_{SIR} values are constrained within a realistic range, maintaining consistency with the underlying biological principles and ensuring computational stability within differential equation-based epidemic simulations.

Step 1: ζ_{max} Network Scaling Calculation

The initial step computes the maximum possible zeta value for network normalization:

$$\zeta_{\text{max}} = n \times \zeta_{\text{mean}} \times (1 + \delta_{\text{sensitivity}})$$

Where:

- $n = \text{len}(zeta_nodes)$ is the number of active symptoms,
- ζ_{mean} is the average zeta value across all active symptoms,
- $\delta_{\text{sensitivity}}$ is the network sensitivity parameter.

This computed ζ_{max} serves as a normalization upper bound for subsequent zeta-based transformations and ensures consistent scaling across varying symptom loads and network conditions.

Step 2: Adaptive Critical Threshold Determination

Critical threshold values (ζ_{crit}) are dynamically assigned based on the number of active symptoms:

$$\zeta_{\text{crit}} = \begin{cases} 0.15 & \text{if } n \leq 2 \quad \text{(mild illness)} \\ 0.20 & \text{if } 3 \leq n \leq 4 \quad \text{(moderate illness)} \\ 0.30 & \text{if } 5 \leq n \leq 6 \quad \text{(severe illness)} \\ 0.40 & \text{if } n \geq 7 \quad \text{(critical illness)} \end{cases}$$

These thresholds enable accurate adjustment of recovery rates and capture shifts in disease severity within the epidemiological model.

Step 3: Network-Scaled ζ Calculation

Raw zeta values are transformed through sigmoid scaling:

$$\zeta_{\text{net}} = \frac{\zeta_{\text{max}}}{1 + e^{-k_{\text{sigmoid}} \times (\zeta_{\text{raw}} - \zeta_{\text{crit}})}}$$

Where:

- $k_{\text{sigmoid}} = 4$ controls the steepness of the transformation.

This transformation ensures that raw zeta values are smoothly normalized into network-scaled severity metrics, maintaining biological plausibility and mathematical tractability.

Core γ Calculation

The core gamma calculation employs an inverse sigmoid transformation:

$$\gamma = \frac{1}{1 + e^{+k_{\text{sigmoid}} \times (\zeta_{\text{net}} - \zeta_{\text{crit}})}}$$

This formula transforms the network-scaled zeta values into biologically plausible recovery rates for epidemiological modeling.

2.1.6 Calculation of β^+

The calculation of the B^+ value considers the physical characteristics of symptoms within their respective transmission mechanisms.

Sneezing

The transmission potential of sneezing has been evaluated using a multi-parameter weighted coefficient analysis based on its physical characteristics within transmission mechanisms. The parameters considered include: P (spectrum breadth), T (airborne persistence), S (sedimentation rate), C

(pathogen concentration), and D (long-range dispersal). Each parameter is weighted according to its relative importance in transmission dynamics, and combined using the following equation:

$$\beta^+ = w_P \cdot P + w_T \cdot T + w_S \cdot (1 - S) + w_C \cdot C + w_D \cdot D$$

In this formulation, the weights are defined as:

$$w_P = 0.15, \quad w_T = 0.2, \quad w_S = 0.15, \quad w_C = 0.25, \quad w_D = 0.25$$

The corresponding parameter values for sneezing are:

$$P = 1.0, \quad T = 0.8, \quad S = 0.6, \quad C = 0.9, \quad D = 1.0$$

The sedimentation rate is used as $1 - S$ in the equation to reflect that lower sedimentation implies longer airborne duration, thereby increasing transmission risk. Substituting into the equation yields:

$$\beta^+ = 0.15 \cdot 1.0 + 0.2 \cdot 0.8 + 0.15 \cdot (1 - 0.6) + 0.25 \cdot 0.9 + 0.25 \cdot 1.0 = 0.845$$

As a result, the recommended value of β^+ for sneezing is approximately $\boxed{0.845}$. This high value indicates that sneezing is an extremely potent transmission vector due to its ability to project pathogen-laden particles over long distances and its association with high viral loads.

Cough

The transmission potential of dry cough has been evaluated through a weighted coefficient analysis based on its physical characteristics within respiratory transmission mechanisms. The parameters considered include: P (particle size), T (airborne persistence), S (sedimentation rate), C (pathogen concentration), and D (dispersal distance). Each parameter is weighted according to its contribution to transmission dynamics and combined through the following equation:

$$\beta^+ = w_P \cdot P + w_T \cdot T + w_S \cdot (1 - S) + w_C \cdot C + w_D \cdot D$$

The weights applied in this formulation are:

$$w_P = 0.15, \quad w_T = 0.2, \quad w_S = 0.15, \quad w_C = 0.25, \quad w_D = 0.25$$

The normalized parameter values for dry cough are:

$$P = 0.9, \quad T = 0.7, \quad S = 0.6, \quad C = 0.7, \quad D = 0.7$$

Sedimentation is modeled inversely as $1 - S$ to reflect that lower sedimentation increases airborne duration. Substituting the values into the equation yields:

$$\beta^+ = 0.15 \cdot 0.9 + 0.2 \cdot 0.7 + 0.15 \cdot (1 - 0.6) + 0.25 \cdot 0.7 + 0.25 \cdot 0.7$$

$$\beta^+ = 0.135 + 0.14 + 0.06 + 0.175 + 0.175 = \boxed{0.685}$$

As a result, the recommended β^+ value for dry cough is approximately **0.685**. While this is lower than sneezing ($\beta^+ = 0.845$), it still reflects a relatively high transmission potential. This is primarily due to the efficient production of aerosols, which can remain airborne for extended periods. Additionally, the small particle size associated with dry cough facilitates deeper lung penetration, thereby increasing infection risk despite a comparatively moderate viral load.

Nasal Discharge

The contact transmission potential of nasal discharge is evaluated based on three key parameters: surface viral load, contact surface area, and virus transfer efficiency. These are integrated into the following weighted formula:

$$\beta_{\text{contact}} = w_1 \cdot \tilde{C}_v + w_2 \cdot \tilde{A} + w_3 \cdot \tilde{P}_{\text{transfer}}$$

Where:

- $\tilde{C}_v$: Normalized surface viral concentration (range: 0–1),
- $\tilde{A}$: Normalized contact surface area,

- $\tilde{P}_{\text{transfer}}$: Normalized virus transfer rate,
- w_1, w_2, w_3: Weight coefficients for each parameter, with $w_1 + w_2 + w_3 = 1$.

Parameter Normalization

- Measured surface viral load: $C_v = 330\,\text{virion/cm}^2$
  Normalized: $\tilde{C}_v = \frac{330}{1000} = 0.33$
- Contact area: $A = 150\,\text{cm}^2$
  Normalized: $\tilde{A} = \frac{150}{200} = 0.75$
- Transfer rate: $P_{\text{transfer}} = 0.2$ (already normalized)

Weight Assignment (Literature- and Intuition-Based)

$$w_1 = 0.4, \quad w_2 = 0.3, \quad w_3 = 0.3$$

Final Calculation

$$\beta_{\text{contact}} = 0.4 \cdot 0.33 + 0.3 \cdot 0.75 + 0.3 \cdot 0.2 = 0.132 + 0.225 + 0.06 = \boxed{0.417}$$

Conclusion
The normalized, weighted contact transmission coefficient for nasal discharge is:

$$\beta^{+}_{\text{nasal}} = 0.417$$

This value indicates a moderate level of transmission risk through surface contact. Given the considerable viral load, reasonably large contact area, and measurable transfer potential, nasal discharge poses a non-negligible risk of indirect transmission via contaminated surfaces. This scenario is particularly relevant in environments involving shared surfaces such as door handles, desks, or public equipment.

Diarrhea

The fecal-oral transmission potential of rotavirus-associated diarrhea can be quantified based on physical parameters including viral concentration in feces, surface persistence, transfer efficiency, and contact area. The normalized transmission coefficient is calculated using the following weighted equation:

$$\beta_{\text{fecal}} = w_1 \cdot \tilde{C}_v + w_2 \cdot \tilde{T}_s + w_3 \cdot \tilde{\eta} + w_4 \cdot \tilde{A}$$

Parameter Definitions

- $\tilde{C}_v$: Normalized viral load in feces (reference maximum: 10^{12} virions/g, typical for norovirus)
- $\tilde{T}_s$: Normalized surface persistence duration (reference maximum: 14 days)
- $\tilde{\eta}$: Normalized fecal-hand-mouth transfer efficiency
- $\tilde{A}$: Normalized contaminated surface area (reference maximum: 200 cm^2) — not included in this calculation

Normalized Input Values (for Rotavirus)

- Viral load: 10^{10} virions/g $\Rightarrow \tilde{C}_v = \frac{10^{10}}{10^{12}} = 0.01$
- Surface persistence: approx. 5 days $\Rightarrow \tilde{T}_s = \frac{5}{14} \approx 0.36$
- Transfer efficiency: mean value of 0.4 $\Rightarrow \tilde{\eta} = 0.4$

Weight Assignment

Based on biological relevance and empirical estimation:

$$w_1 = 0.4, \quad w_2 = 0.3, \quad w_3 = 0.3$$

Calculation

$$\beta_{\text{fecal}} = 0.4 \cdot 0.01 + 0.3 \cdot 0.36 + 0.3 \cdot 0.4 = 0.004 + 0.108 + 0.12 = \boxed{0.232}$$

Conclusion

The normalized fecal-oral transmission coefficient for rotavirus-induced diarrhea is:

$$\beta^{+}_{\text{fecal}} = 0.232$$

This value indicates a moderate transmission risk via fecal contamination. Although rotavirus has a lower environmental persistence and viral load compared to norovirus, its high infectivity and the possibility of fecal-hand-mouth contact under poor hygiene conditions make it a significant public health concern. This model provides a practical risk assessment based on physical and environmental parameters involved in indirect transmission pathways.

Pneumonia

Pneumonia is classified by the World Health Organization (WHO) and the American Thoracic Society/Infectious Diseases Society of America (ATS/IDSA) as a severe or very severe disease in both children and adults. While mild-to-moderate cases typically resolve in 2–3 weeks, full recovery may take 1–3 months. Several clinical and systemic factors influence recovery, including complication risk, ICU admission rate, antibiotic resistance, and healthcare accessibility.

To quantify the treatment difficulty, we define the recovery difficulty index ζ using the following inverse-proportional formula:

$$\zeta_s = \frac{1}{Z}\left[w_1 \cdot \text{severity}_s + w_2 \cdot \left(\frac{\text{recovery_time}_s}{t_{\max}}\right) + w_3 \cdot (1 - \text{accessibility}_s)\right]$$

Model Inputs

- Severity score: severity $= 0.9$
- Average recovery time: recovery_time $= 30$ days
- Maximum reference time: $t_{\max} = 60$ days
- Healthcare accessibility score: accessibility $= 0.5$

Weights

$$w_1 = 0.4 \quad (\textit{severity}), \quad w_2 = 0.4 \quad (\textit{recovery duration}), \quad w_3 = 0.2 \quad (\textit{treatment accessibility})$$

Calculation

$$\zeta = \frac{1}{1}\left[0.4 \cdot 0.9 + 0.4 \cdot \left(\frac{30}{60}\right) + 0.2 \cdot (1 - 0.5)\right] = 0.36 + 0.2 + 0.1 = \boxed{0.66}$$

Based on the above parameters, the calculated recovery difficulty score for pneumonia is $\zeta = 0.66$. This indicates that pneumonia falls into the **moderate-to-high** recovery difficulty range. While it involves clinically severe symptoms, its moderate recovery time and partial treatment accessibility prevent it from being classified as the most difficult to treat. In low-resource settings, however, reduced accessibility may increase the ζ score significantly, elevating pneumonia into a high-risk treatment category.

Sepsis

Sepsis is a life-threatening condition characterized by systemic inflammation and organ dysfunction, typically associated with a high mortality rate. Clinical management often involves intensive care and extended recovery periods, especially in moderate to severe cases. Due to its critical nature and resource-intensive treatment, sepsis represents a high-burden condition for healthcare systems.

To evaluate the treatment difficulty, we compute the recovery difficulty score ζ using the following inverse-weighted formula:

$$\zeta_s = \frac{1}{Z}\left[w_1 \cdot \text{severity}_s + w_2 \cdot \left(\frac{\text{recovery_time}_s}{t_{\max}}\right) + w_3 \cdot (1 - \text{accessibility}_s)\right]$$

Model Inputs

- Severity score: $\text{severity}_s = 0.95$
- Recovery time: $\text{recovery_time}_s = 45$ days
- Maximum reference time: $t_{\max} = 60$ days
- Treatment accessibility score: $\text{accessibility}_s = 0.6$

Weight Assignment

$$w_1 = 0.4, \quad w_2 = 0.4, \quad w_3 = 0.2 \quad \Rightarrow \quad Z = w_1 + w_2 + w_3 = 1.0$$

Calculation

$$\zeta = \frac{1}{1.0}\left[0.4 \cdot 0.95 + 0.4 \cdot \left(\frac{45}{60}\right) + 0.2 \cdot (1 - 0.6)\right]$$

$$= 0.38 + 0.30 + 0.08 = \boxed{0.76}$$

The calculated recovery difficulty score for sepsis is:

$$\zeta_{\text{sepsis}} = \boxed{0.76}$$

This score indicates that sepsis represents a condition with **high treatment difficulty**. While partial access to care may exist in developed countries, the clinical severity and complex management requirements elevate the treatment burden. Sepsis should therefore be classified within the upper tier of difficulty in terms of recovery and resource allocation.

High Fever

High fever is a common symptom observed in systemic infections such as COVID-19, malaria (*P. falciparum*), sepsis, typhoid, and influenza. Clinically, it is classified as:

- Moderate fever: 38–39°C
- High fever: 39–41°C
- Hyperpyrexia (emergency): ¿41°C

Although not lethal on its own, high fever can lead to serious complications depending on the underlying disease. These include:

- Dehydration
- Delirium (especially in the elderly)
- Febrile seizures (common in children)
- Organ stress

To quantify its recovery difficulty, the following zeta formula is used:

$$\zeta_s = \frac{1}{Z}\left[w_1 \cdot \text{severity}_s + w_2 \cdot \left(\frac{\text{recovery_time}_s}{t_{\max}}\right) + w_3 \cdot (1 - \text{accessibility}_s)\right]$$

Model Inputs

- $\text{severity}_s = 0.75$

- $\text{recovery_time}_s = 10$ days
- $\text{accessibility}_s = 0.85$
- $t_{\max} = 60$ days (reference for scaling)

Weights

$$w_1 = 0.4, \quad w_2 = 0.4, \quad w_3 = 0.2, \quad Z = w_1 + w_2 + w_3 = 1.0$$

Calculation

$$\zeta = 0.4 \cdot 0.75 + 0.4 \cdot \left(\frac{10}{60}\right) + 0.2 \cdot (1 - 0.85)$$

$$= 0.3 + 0.0667 + 0.03 = \boxed{0.3967}$$

Conclusion
The calculated recovery difficulty score for high fever is:

$$\zeta_{\text{fever}} = \boxed{0.40}$$

This value suggests that while fever is not deadly by itself, it can cause significant clinical difficulty when compounded by underlying diseases. With early diagnosis and symptomatic management, it can typically be controlled, but in critical conditions it may contribute to increased treatment complexity.

Loss of Consciousness

Loss of consciousness is a severe neurological symptom that may arise during systemic infections such as sepsis, encephalitis, or meningitis. Clinically, it is often associated with coma, multi-organ failure, and high mortality. Intensive care is frequently required, and recovery may take weeks or even months. Long-term neurological sequelae are not uncommon. Treatment access varies significantly by region, depending on the availability of antibiotics, antivirals, and intensive care units.

Model Inputs

- Severity score: 0.98
  Justification: One of the most severe symptoms; associated with coma, organ failure, and high mortality.
- Recovery time: 45 days
  Justification: Cognitive recovery after sepsis or encephalitis may take weeks to months.
- Accessibility: 0.5
  Justification: Access to treatment varies by region (e.g., antibiotics, antivirals, ICU).

Weight Assignment

$$w_1 = 0.4, \quad w_2 = 0.4, \quad w_3 = 0.2 \quad \Rightarrow \quad Z = w_1 + w_2 + w_3 = 1.0$$

Calculation

$$\zeta = 0.4 \cdot 0.98 + 0.4 \cdot \left(\frac{45}{60}\right) + 0.2 \cdot (1 - 0.5)$$

$$= 0.392 + 0.3 + 0.1 = \boxed{0.792}$$

The calculated recovery difficulty score for infection-related loss of consciousness is:

$$\zeta_{\text{consciousness}} = \boxed{0.79}$$

This score indicates that loss of consciousness is a very high severity symptom, requiring long recovery time, and characterized by moderate treatment accessibility. It should be prioritized accordingly in symptom network modeling.

1. Both symptoms must have a defined `beta_raw` value, indicating their potential to contribute to disease transmission.
2. Both symptoms must belong to the same physiological system (i.e., they share the same system label).
3. The symptoms must co-occur within the same daily symptom combination.

The resulting graph is undirected, meaning that a connection from symptom A to symptom B is treated identically to one from B to A. This network structure is subsequently used to calculate the overall transmission potential (`Beta_SIR`) and the severity of illness (`Gamma_SIR`) within the population.

2.2 Design of Agent Based Model

This study presents an agent-based model designed for epidemiological research applications. The model aims to simulate complex epidemiological processes by modeling individual agents' health conditions, social interactions, and behavioral characteristics. The model architecture is designed to encompass each agent's unique immune system capacity (categorized into three levels: strong, normal, and weak), infection status (susceptible, infected, recovered, dead), symptom profiles, and social behavioral patterns. Interactions between agents are modeled on a small-world network topology, incorporating real-life factors such as daily activities (cafeteria usage, social distancing behavior) and healthcare service accessibility (hospital capacity, normal and intensive care units). The model's outputs provide the capacity to analyze population-level epidemiological dynamics through R0 values, infection spread rates, critical intervention points, and various scenario analyses.

2.2.1 Agent Characteristics and Classification

Individual Agent Definition and Basic Parameters

In the agent-based model developed in this study, each agent represents an individual and possesses the following basic characteristics:

- **Unique identification number (ID):** For tracking each agent within the system.
- **Infection status:** Status within the SIRD model framework (Susceptible-Infected-Recovered-Dead).
- **Immune system class:** Individual resistance capacity against disease.
- **Health level:** Health status varying between 0–1000.
- **Network connections:** Neighborhood relationships within the small-world network.
- **Symptom status:** Current symptom profile and severity.

Model Parameterization and User Inputs

The model is configured based on fundamental parameters determined by the user:

- **Total number of agents:** The number of individuals to be included in the simulation is determined by the user.
- **Immune system distribution:** The proportions of individuals with strong, normal, and weak immune systems within the population are defined through user input.
- **Network connection count:** The number of neighbors each agent will have in the small-world network is determined by the user.

Immune System Capacity Classification

The model defines three different immune system capacity classes to reflect population heterogeneity. The distribution of these classes within the population occurs according to proportions determined by the user:

- **Strong Immune System Class**
  Initial health level: Randomly assigned value between 601–1000.
- **Normal Immune System Class**
  Initial health level: Randomly assigned value between 301–600.
- **Weak Immune System Class**
  Initial health level: Randomly assigned value between 100–300.

This classification enhances the model's realism by reflecting immunological heterogeneity in real populations. The incorporation of each class with different initial health reserves creates meaningful differences in disease progression and outcomes.

2.2.2 Disease Dynamics

In this study, disease transmission is simulated through the SIRD (Susceptible-Infected-Recovered-Dead) model. The model assumes that each agent can be in one of four different health states: Susceptible (S) - individuals susceptible to contracting the disease, Infected (I) - individuals who are actively ill and infectious, Recovered (R) - individuals who have overcome the disease and gained immunity, and Dead (D) - individuals who have lost their lives due to the disease. During the infection process, agents undergo dynamic processes such as daily symptom development, changes in health scores, death risk calculations, and recovery probability assessments. These state transitions occur with different probabilities depending on the agents' immune system classes, current health scores, and symptom severity, thereby achieving realistic epidemiological dynamics.

2.2.3 Infection Probability

Susceptible-Infected (S-I) Transition Mechanism

In this model, the S-I transition is performed through a complex probabilistic calculation. The transition process is shaped around the following fundamental equation:

Basic Infection Probability Formula

For each susceptible (S) agent, the infection probability is calculated using the formula below:

$$p = 1 - \exp\left(-\beta_i \times \frac{N_i}{d_i}\right)$$

Formula Components:

- p: Infection probability
- β_i: Infectiousness parameter of the infected agent
- N_i: Number of infected neighbors of the susceptible agent
- d_i: Total number of neighbors (network degree) of the susceptible agent

Immune System Modification

The calculated infection probability is modified based on the immune system capacity of the susceptible agent:

$$p_{\text{final}} = p \times \sigma_i$$

Immune System Multipliers (σ_i):

- Strong immune system: $\sigma_i = 0.8$ (20% lower infection risk)
- Normal immune system: $\sigma_i = 1.0$ (standard risk)
- Weak immune system: $\sigma_i = 1.4$ (40% higher infection risk)

Network-Based Interaction Model

The model combines small-world network topology with cafeteria interactions to form realistic contact patterns.

Total Neighbor List:

$$\text{Neighbors}_{\text{total}} = \text{Neighbors}_{\text{network}} \cup \text{Neighbors}_{\text{cafeteria}}$$

For each susceptible agent:

- Direct neighbors in the network are identified
- Agents sharing the same cafeteria table are added
- For each infected agent in the combined neighbor list, the infection probability is calculated using the formula above

Daily Infection Process

Each simulation day consists of the following steps:

- **Neighbor Detection**: The active neighbor list is formed for each susceptible agent
- **Beta Value Retrieval**: The current β value is determined based on the infected agent's illness progression
- **Probability Calculation**: Infection probability is computed using the base formula
- **Immune System Effect**: The probability is modified based on the susceptible agent's immune capacity
- **Stochastic Decision**: A Bernoulli trial is conducted based on the final probability to determine whether infection occurs

Dynamic Parameter Update

The β values are updated daily based on the infected agent's disease progression. These updates depend on:

- Time elapsed since the agent was infected
- Predefined symptom data table
- Symptom weighting and physiological system interactions

Infection Realization

If the Bernoulli trial yields a positive outcome, the susceptible agent transitions to the infected status, and the state changes from 'S' to 'I'. Initial infection parameters are assigned.

This comprehensive model simulates realistic S-I transitions by incorporating heterogeneity in epidemiological processes, network effects, and individual immunological variations.

2.2.4 Infection Dynamics and Recovery Probability

1. Basic Recovery Probability

Recovery probability is calculated using an exponential decay model:

$$P_{\text{recovery}}(t) = 1 - e^{-\max(\gamma(t), 0.01)}$$

Where:

- $\gamma(t)$: Daily recovery rate at time t
- 0.01: Minimum recovery rate (lower bound)

2. Critical Threshold Assignment
The disease severity threshold ζ_{crit} is determined based on the average severity ζ_{mean}:

$$\zeta_{\text{crit}} = \begin{cases} 0.15, & \text{if } \zeta_{\text{mean}} < 0.03 \\ 0.20, & \text{if } 0.03 \leq \zeta_{\text{mean}} < 0.08 \\ 0.30, & \text{if } 0.08 \leq \zeta_{\text{mean}} < 0.20 \\ 0.40, & \text{if } \zeta_{\text{mean}} \geq 0.20 \end{cases}$$

3. Immune System Modification
Recovery probability is scaled by individual immune system capacity:

$$P_{\text{recovery_modified}} = P_{\text{recovery}} \times M_{\text{immunity}}$$

Where:

$$M_{\text{immunity}} = \begin{cases} 1.5, & \text{Strong immune system} \\ 1.0, & \text{Normal immune system} \\ 0.6, & \text{Weak immune system} \end{cases}$$

4. Health Reserve Bonus
Final recovery probability is further scaled by a health bonus factor depending on health score and immunity class:

$$P_{\text{final}} = P_{\text{recovery_modified}} \times B_{\text{health}}(H_{\text{score}}, I_{\text{class}})$$

For Strong Immunity:

$$B_{\text{health}} = \begin{cases} 4.0, & \text{if } H_{\text{score}} > 700 \\ 2.5, & \text{if } 500 < H_{\text{score}} \leq 700 \\ 1.5, & \text{if } 300 < H_{\text{score}} \leq 500 \\ 0.8, & \text{if } H_{\text{score}} \leq 300 \end{cases}$$

For Normal Immunity:

$$B_{\text{health}} = \begin{cases} 3.0, & \text{if } H_{\text{score}} > 450 \\ 2.0, & \text{if } 300 < H_{\text{score}} \leq 450 \\ 1.0, & \text{if } 150 < H_{\text{score}} \leq 300 \\ 0.6, & \text{if } H_{\text{score}} \leq 150 \end{cases}$$

For Weak Immunity:

$$B_{\text{health}} = \begin{cases} 2.0, & \text{if } H_{\text{score}} > 200 \\ 1.2, & \text{if } 120 < H_{\text{score}} \leq 200 \\ 0.8, & \text{if } 80 < H_{\text{score}} \leq 120 \\ 0.4, & \text{if } H_{\text{score}} \leq 80 \end{cases}$$

5. Two-Step Recovery Condition
Recovery occurs only if both conditions are met:
5.1 Probabilistic Condition:

$$U \sim \text{Uniform}(0, 1), \quad U < P_{\text{final}}$$

5.2 Cumulative Recovery Condition:

$$R_{\text{score}}(t) = \sum_{\tau=0}^{t} \gamma(\tau), \quad R_{\text{score}}(t) \geq R_{\text{threshold}}$$

Where $R_{\text{threshold}}$ is drawn from:

$$R_{\text{threshold}} \sim \text{Uniform}(1.5, 3.5)$$

6. State Transition Function

$$\text{Status}(t+1) = \begin{cases} 'D', & \text{if } U_{\text{death}} < P_{\text{death}}(H_{\text{score}}(t)) \\ 'R', & \text{if } U_{\text{recovery}} < P_{\text{final}} \text{ AND } R_{\text{score}}(t) \geq R_{\text{threshold}} \\ 'I', & \text{otherwise} \end{cases}$$

7. Health Score Dynamics

$$\text{health_score}(t+1) = \text{health_score}(t) - \text{daily_health_loss}$$

Daily Health Loss Calculation:

$$\text{daily_health_loss} = \sum(\text{symptom_loss} \times \text{immunity_modifier})$$

Modifiers based on immune system:

- Strong: 0.7 (30% reduction)
- Normal: 1.0 (standard)
- Weak: 1.3 (30% increase)

8. Additional Health Dynamics

8.1 Hospitalization Bonus (subtracted):

$$\text{if hospitalized:} \begin{cases} -18, & \text{pandemic hospital} \\ -25, & \text{ICU} \\ -12, & \text{regular unit} \end{cases}$$

8.2 Natural Recovery Bonus (home treatment):

$$\text{if not hospitalized:} \quad \text{daily_health_loss} -= 4.0$$

8.3 Final Calculation:

$$\text{daily_health_loss} = \max(0, \text{daily_health_loss})$$

$$\text{health_score}(t+1) = \max(0, \text{health_score}(t) - \text{daily_health_loss})$$

Initial Health Score Based on Immunity Class:

- Strong Immunity: 601–1000
- Normal Immunity: 301–600
- Weak Immunity: 100–300

This framework ensures different individuals start with different reserves and lose health at varying rates depending on symptoms and treatment conditions.

2.2.5 Dynamics of Death

This epidemiological model differs from traditional models by utilizing a health-based dynamic mortality system.

Physiological System Synergy and Mortality Dynamics

This model calculates the interaction between different physiological systems using a synergy mechanism. A synergy coefficient of 1.0 (100%) is assigned for symptoms within the same system, 0.8 (80%) for combinations such as the nervous and respiratory systems, 0.6 (60%) for immune-respiratory and immune-nervous combinations, and 0.2 (20%) for other system pairings. The model uses the following formula:

$$\text{Total Severity} = \text{Raw Severity} + \text{Synergy Severity}$$

Synergy severity is computed using the following weights:

- w_{sys}: Physiological system weight,
- w_{net}: Network distance weight,
- w_{ij}: Combination-specific weight.

Regarding direct effects on mortality, single symptoms cause a specific amount of daily health loss. For instance:

- Cough: 0.5 health loss per day,
- Fever: 0.75 health loss per day.

However, certain combinations are modeled as highly lethal:

- **Pneumonia + High Fever**: considered deadly when co-occurring,
- **Sepsis + Coma**: defined as the most dangerous combination due to maximal synergy,
- **Respiratory + Nervous system combinations**: result in 80% additional severity.

The model explicitly defines the following as **critical combinations**:

1. Respiratory + Nervous system (e.g., Pneumonia + Coma),
2. Immune + Respiratory system (e.g., Sepsis + Pneumonia),
3. Multiple symptoms within the same system (e.g., Cough + Runny nose + Pneumonia).

Furthermore, the model dynamically adjusts the recovery rate γ_{sir} based on symptom severity:

- For mild symptoms: $\gamma_{\text{sir}} = 0.5$ (fast recovery),
- For multiple severe symptoms: $\gamma_{\text{sir}} < 0.3$ (very slow recovery).

Thanks to this sophisticated structure, the co-occurrence of symptoms is not merely additive; it is multiplicatively amplified — leading to a more realistic estimation of mortality probability and recovery delays.

2.3 Social Network

Standard Agent Network Topology

Agents are typically organized according to a small-world network model. The user can define key parameters such as the total number of agents, the rewiring probability, and the number of neighbors each agent is connected to. The reason for selecting the small-world network model is its ability to realistically represent communication patterns in a society using a simplified structural framework [MN00].

Epidemic Day Network Topology

The day on which the hospital discharge rate falls below the hospital admission rate is declared as the **epidemic day**. On this day, the social network transitions into a **clique structure** composed of 0 to 3 agents. This clique configuration is designed to represent **social isolation** within the population.

After the normalization day, the network structure transitions back to a small-world topology, restoring regular social connectivity among agents.

2.4 Locations

2.4.1 Hospital and Pandemic Hospital

Hospitals consist of two main units: the intensive care unit (ICU) and the general care unit. The assignment of patients to these units depends on the severity of symptoms, immune system classification (strong, normal, weak), and the individual's health score. Patients showing critical symptoms such as Coma, Sepsis, or Loss of Consciousness are directly admitted to the ICU. Those with severe symptoms such as Pneumonia or High Fever are also directed to the ICU if their health score is low based on immune status. All other patients are admitted to the general care unit.

The total hospital bed capacity and the occupancy rate of each unit are defined by the user. The system supports dynamic adjustments depending on the epidemic status. During the normal period, hospitals operate with standard capacity. After the epidemic day is declared, hospital capacity increases from 100% to 120%. At this stage, the user can also choose to activate a pandemic hospital. Additionally, patients in critical condition are automatically transferred from general care to ICU if necessary. This infrastructure enables the healthcare system to be dynamically optimized based on evolving conditions.

2.4.2 Cafeteria

The number of tables and the seating capacity per table are defined by the user. Each table forms a clique structure, meaning that all individuals at the same table are fully connected to each other. There is no social interaction between different tables, effectively modeling isolated social clusters within the cafeteria environment.

Hunger System

1. Hunger Parameters

- **Daily Hunger Increment:** 0.5 units
- **Hunger Threshold:** 2.0 units

2. Hunger Clock System

Initialization:

- Each agent is assigned a random initial hunger level: 0.0, 0.5, or 1.0
- This ensures agents become hungry at different times.

Daily Update:

- Each day, the hunger clock increases by 0.5 units.

3. Cafeteria Visit Decision Algorithm

1. **Critical Symptom Check:** If the agent is infected and has critical symptoms (e.g., coma, sepsis, loss of consciousness), they are strictly prohibited from visiting the cafeteria.
2. **Hunger Threshold Check:** If the hunger clock exceeds the threshold (2.0), the agent will definitely go to the cafeteria.
3. **Probability-Based Decision:** Otherwise, the probability of visiting increases with hunger and is calculated as the ratio of current hunger to the threshold.

4. Hunger-Based Decision Table

Hunger Level	Probability to Visit	Explanation
0.0 – 0.5	0–25%	Not hungry
0.5 – 1.0	25–50%	Mild hunger
1.0 – 1.5	50–75%	Moderate hunger
1.5 – 2.0	75–100%	High hunger
$\geq$ 2.0	100%	Extremely hungry – definitely visits

5. Cafeteria Group Formation System
Application Rules:

- Only agents who are not hospitalized can apply.
- Only agents in states `S`, `I`, or `R` are eligible.
- Decision to apply is determined by the hunger-based cafeteria decision function.

Seating:

- Tables are filled randomly, within a predefined maximum seat capacity.

6. Symptom-Based Constraints
Agents with Critical Symptoms:

- Cannot visit the cafeteria under any condition.
- Cannot establish any social connections.
- Examples: Coma, Sepsis, Loss of Consciousness

Agents with B^- Symptoms:

- Can visit the cafeteria but have limited interaction.
- Maximum of 3 social connections allowed.
- Examples: Pneumonia, High Fever, etc.

7. Eating and Hunger Reset

- After eating, an agent's hunger clock is reset to 0.0.

8. Hunger Cycle Example

- Day 1: Hunger = 0.5 $\rightarrow$ 25% chance $\rightarrow$ May not go
- Day 2: Hunger = 1.0 $\rightarrow$ 50% chance $\rightarrow$ May or may not go
- Day 3: Hunger = 1.5 $\rightarrow$ 75% chance $\rightarrow$ Likely to go
- Day 4: Hunger = 2.0 $\rightarrow$ 100% chance $\rightarrow$ Definitely goes

9. Social Interaction Rules
At the Table:

- All agents at the same table become connected.
- Agents with critical symptoms are excluded.
- B^- agents can only connect with a maximum of 3 others.
- Social links carry infection risk.

Cafeteria Analysis:

- Daily visitor counts are recorded.
- Symptom distributions are analyzed.
- Compared with hospital occupancy rates.

2.5 Daily Symptom Graph Construction

In this model, a symptom graph is constructed for each day based on the observed symptoms. Each symptom that appears on a given day is represented as a node in the graph. Edges between nodes (i.e., symptoms) are formed only when the following three conditions are simultaneously satisfied:

1. Both symptoms must have a defined βraw value, indicating their potential to contribute to disease transmission.
2. Both symptoms must belong to the same physiological system (i.e., they share the same system label).
3. The symptoms must co-occur within the same daily symptom combination.

The resulting graph is undirected, meaning that a connection from symptom A to symptom B is treated identically to one from B to A. This network structure is subsequently used to calculate the overall transmission potential (`Beta_SIR`) and the severity of illness (`Gamma_SIR`) within the population.

3 Scenarios

3.1 Upper Tract Respiratory Infection

Upper respiratory tract infections (URTIs) are commonly observed viral illnesses characterized by symptoms such as sneezing, coughing, runny nose, muscle pain, and fever. These infections, while often self-limiting and non-lethal, can exhibit significant transmission dynamics in densely connected populations, particularly in communal areas such as cafeterias or dormitories.

In this simulation scenario, we model the spread of an URTI using a symptom-based agent framework, where each agent's contagiousness, disease severity, and recovery rate are determined by their symptom profile and immune status. The virtual population is composed of agents with varying immunity levels—30% strong, 60% normal, and 10% weak—allowing us to capture heterogeneity in disease progression and containment.

This setup enables a detailed exploration of how symptom-driven transmission interacts with immune distribution and behavioral patterns to shape the overall course of an outbreak.

Table 2: Daily symptom profile and transmission parameters

Day	**Symptoms**	$\beta_{combined}$	β_{SIR}	ζ_{raw}	ζ_{net}	γ_{SIR}
0	Runny Nose, Sneezing, Muscle Pain	1.8000	0.6429	0.988	0.8218	0.4000
1	Runny Nose, Sneezing, Muscle Pain	1.8000	0.6429	0.988	0.8218	0.4000
2	Runny Nose, Sneezing, Muscle Pain	1.8000	0.6429	0.988	0.8218	0.4000
3	Cough, Runny Nose, Fever, Muscle Pain	1.6023	0.6157	1.589	1.2493	0.3000
4	Cough, Runny Nose, Fever, Muscle Pain	1.6023	0.6157	1.589	1.2493	0.3000
5	Muscle Pain	0.0000	0.0000	0.350	0.1891	0.6992

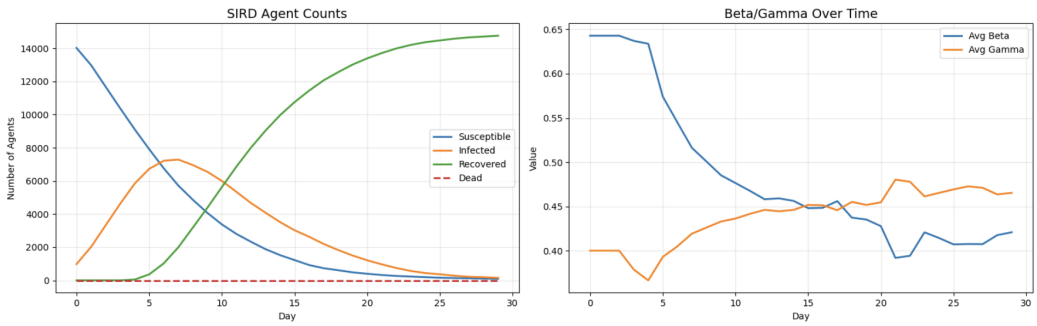


Figure 1: SIRD Graph of Upper Tract Resporitory Infection and Average β and γ values.

The SIRD curves in the left panel of Figure 1 show that in the early phase of the simulation the infection spreads rapidly—as evidenced by the steep rise of the *Infected* curve—peaking at approximately day 7–8. Following this peak, the number of infected agents declines sharply, and by day 30 nearly all agents have recovered (*Recovered* $\approx 14\,500$). The number of deaths remains virtually zero, confirming the model's low-fatality assumption. This rapid resolution is largely driven by the timely infection and subsequent quick recovery of the majority of agents with normal and strong immunity (90%).

The right panel of Figure 1 illustrates that the average transmission rate, β, initially stabilizes around 0.64, then declines in parallel with changes in symptom profiles between days 5 and 10, reaching approximately 0.40–0.45 by day 20. Concurrently, the recovery rate, γ, decreases to about 0.37 during the first five days before increasing and stabilizing in the 0.45–0.48 range. The critical threshold $\beta < \gamma$ is crossed around day 15, marking the onset of the epidemic's decline.

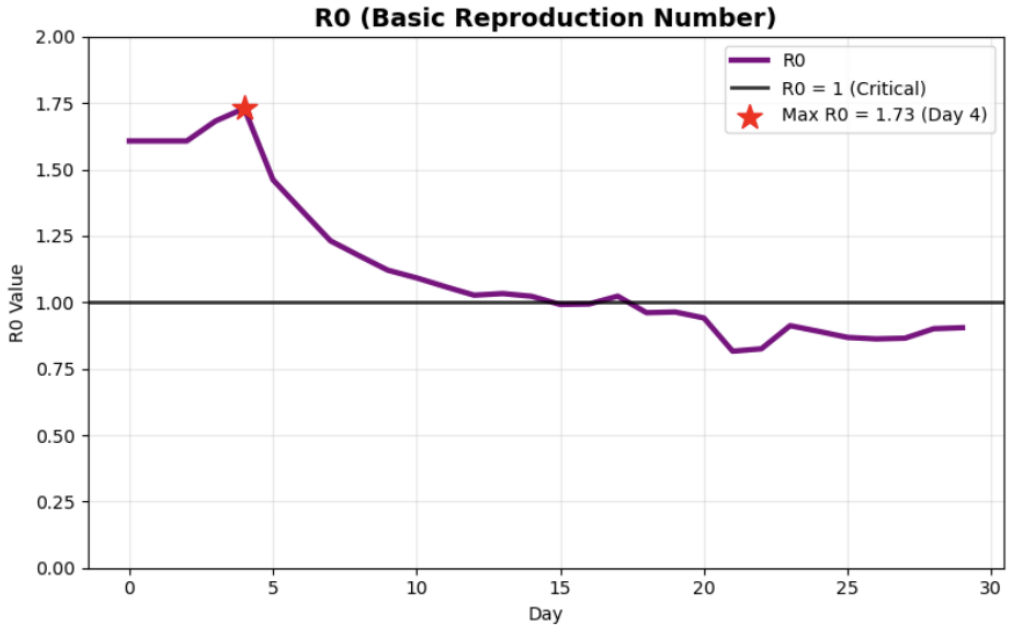


Figure 2: R0 values per day.

As shown in the figure, the basic reproduction number R_0, which begins at approximately 1.6 during the initial days of the simulation, reaches its peak value of 1.73 on day 4—indicating a zenith in daily transmission during the early phase; thereafter, between days 5 and 14, changes in symptom profiles (a reduction in highly transmissible symptoms such as sneezing and runny nose alongside an increase in muscle pain) combined with a rising recovery rate drive R_0 rapidly below 1, surpassing the epidemic's attenuation threshold and ensuring that each infected individual generates fewer than one new case. During days 15–20, R_0 remains roughly stable around the 1.0 threshold, with a transient dip to approximately 0.8 around day 20 followed by a slight rebound to about 0.9, signaling that the small subgroup with weak immunity retains a modest transmission potential and thereby introduces minor fluctuations until the epidemic is fully extinguished. Overall, this time series clearly demonstrates that an outbreak initiated by a high transmission risk in the early stage can be brought under control through symptom-based β/γ dynamics and the distribution of population immunity, ultimately subsiding once the $R_0 < 1$ threshold is exceeded.

This figure presents the full 30-day temporal distribution of symptoms between all agents and provides essential information on the progression of the infection: the incidence of symptoms increases dramatically from day 0, culminating in a peak of approximately 22 500–23 000 symptomatic cases per day around days 6-8, marking the period of maximum transmissibility and clinical burden. At the apex, muscle pain (green) predominates, followed by runny nose (red) and sneezing (purple), whereas fever (orange) and cough (blue) remain comparatively infrequent—an observation that reflects the symptom-based weighting of transmission coefficients (β) in the model whereby muscle pain, despite its lower per-case transmissibility, contributes most substantially to the aggregate symptom count due

to its prevalence. During days 2–4, the daily addition of roughly 3 000–5 000 new symptomatic cases corresponds to an early-phase reproduction number $R_0 > 1$, consistent with rapid outbreak growth. After day 9, symptom counts enter a sustained and pronounced decline, falling below 10 000 cases per day after day 15 and beneath 2 000 cases per day after day 20; this downturn coincides with the point at which the recovery rate (γ) surpasses the transmission rate (β), driving R_0 below unity and signaling epidemic decay. Overall, these results demonstrate that the evolving composition of symptoms exerts a decisive influence on transmission dynamics: highly contagious upper respiratory symptoms dominate initially, but as lower-transmission manifestations such as muscle pain become more prevalent, the epidemic's momentum diminishes, thereby accelerating its resolution.

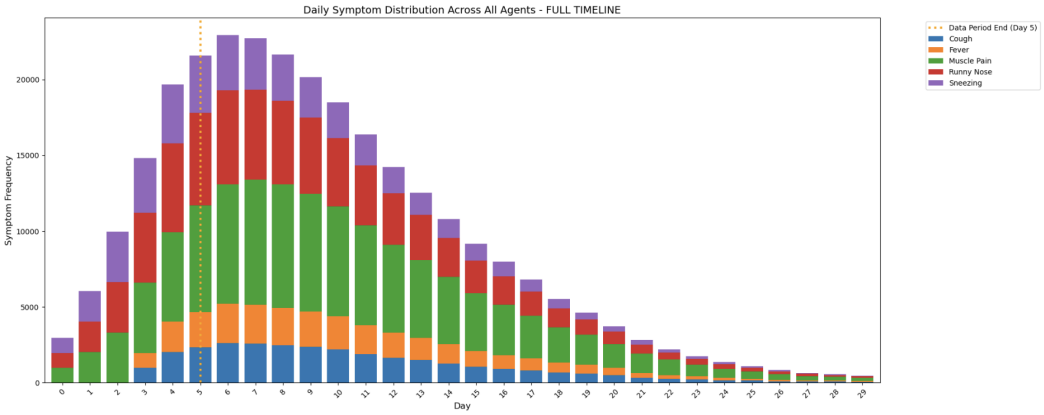


Figure 3: Symptom Frequency per Day.

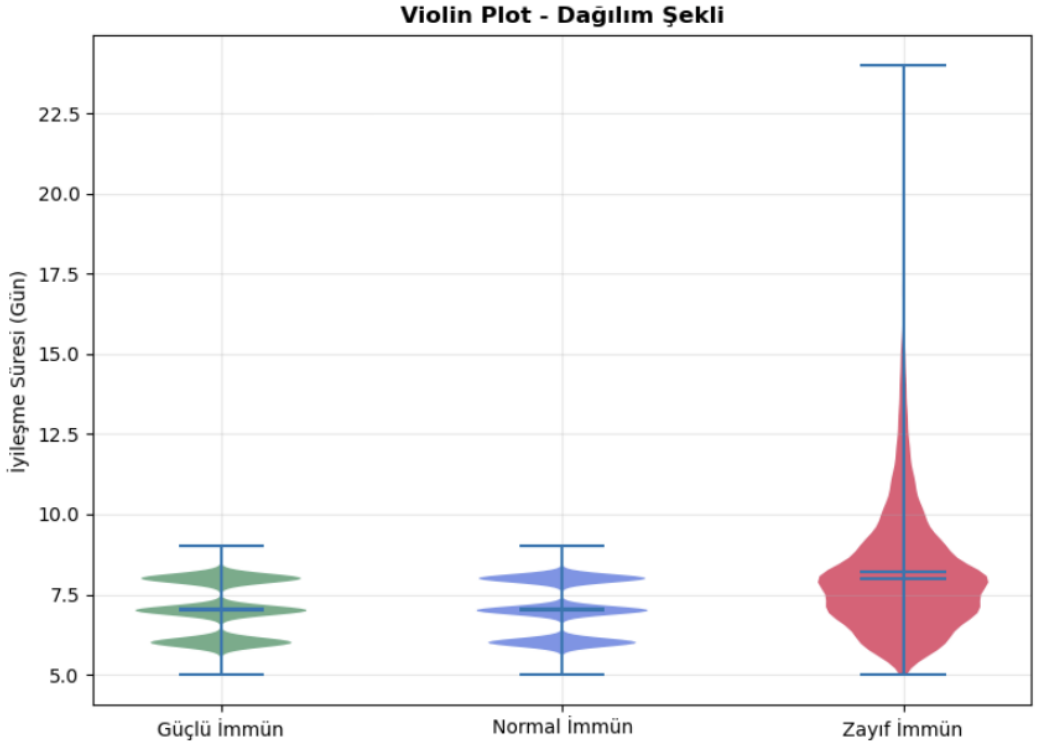


Figure 4: Immunology based healing rate.

This violin plot displays the distribution of recovery times (in days) for the three immunity groups: Strong Immunity (green): The distribution is narrow and symmetric, with most agents recovering

between 6 and 8 days. The median is around 7 days, and the full range spans roughly 5 to 9 days. This tight profile indicates that strong immunity leads to consistently rapid recovery. Normal Immunity (blue): Exhibits a similar distribution to the strong group, with a median of about 7–8 days and a range of approximately 5–9 days. The slightly more compressed whiskers suggest that normal immunity also yields predictable recovery times, though with marginally less consistency than the strong group. Weak Immunity (red): The distribution is both much wider and skewed: recovery times extend from 5 up to 24 days. While the median remains near 8 days, the long upper tail implies that weakly immune individuals often experience significantly prolonged recoveries.

Overall Insight: Strong and normal immunity groups resolve the infection quickly and predictably, whereas the weak immunity subgroup not only introduces considerable variability in recovery duration but can substantially prolong the outbreak's tail. This highlights how a small but vulnerable subpopulation can markedly influence the epidemic's overall dynamics.

3.2 Respiratory Infection

In simulation of a lower respiratory tract infection (LRTI) simulation, we divide a closed population into three immunity groups—40% strong, 50% normal, and 10% weak—while retaining the full set of characteristic symptoms (coughing, sneezing, fever, high fever, muscle pain, and pneumonia). Within a symptom-based agent-model framework, each agent's transmission potential (β), disease severity (ζ), and recovery rate (γ) are dynamically determined by their individual symptom profile and immune status. This setup allows us to isolate the effects of immune heterogeneity on the onset, peak burden, and resolution of the epidemic within a single, unified simulation scenario. Compared to the first scenario, this represents a much more severe illness, and these two scenarios can help us compare diseases characterized by low-severity versus high-severity symptoms.

Table 3: Symptom-based transmission parameters for the LRTI simulation (scenario 2, first 7 days)

Day	Symptoms	β_{combined}	β_{SIR}	ζ_{raw}	ζ_{net}	γ_{SIR}
0	Cough, Sneezing, Fever	2.1888	0.6864	1.0920	0.7904	0.4000
1	Cough, Sneezing, Fever	2.1888	0.6864	1.0920	0.7904	0.4000
2	Cough, Sneezing, Fever	2.1888	0.6864	1.0920	0.7904	0.4000
3	Sneezing, Cough	2.0976	0.6772	0.5400	0.3055	0.5934
4	Sneezing, Cough	2.0976	0.6772	0.5400	0.3055	0.5934
5	Sneezing, Cough	2.0976	0.6772	0.5400	0.3055	0.5934
6	Pneumonia, Cough, Sneezing, Muscle Pain, High Fever	0.0000	0.0021	3.7336	2.1720	0.0008

First Simulation for Respiratory Infection

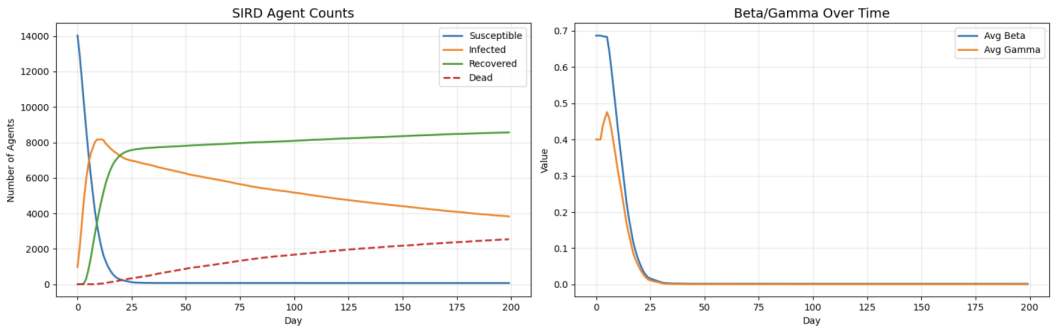


Figure 5: SIRD Graph and Average β and γ .

SIRD Curves (Left Panel) The susceptible population (blue curve) starts above 10 000 on day 0 and declines to nearly zero within the first 20 days, reflecting rapid depletion of susceptibles driven by high initial transmission rates and the early infection of the 10% weak-immunity subgroup. The infected

curve (orange) peaks around days 7–8 at approximately 8 200 agents, after which it enters a steady but gradual decline. Concurrently, the recovered population (green) rises sharply between days 10 and 15 and eventually plateaus at over 8 500 agents, indicating that the majority of infections resolve through recovery. Mortality (red dashed line), while non-negligible, remains limited in scale: by day 200, cumulative deaths approach roughly 2 500, underscoring that even a small vulnerable cohort can contribute to an extended tail of disease-related fatalities despite overall low mortality assumptions.

β/γ Time Series (Right Panel) The average transmission rate, β, begins at around 0.69 during days 0–3, reflecting the predominance of high-transmission symptoms such as coughing, sneezing, and fever. The recovery rate, γ, concurrently rises from approximately 0.40 to 0.48 within the first five days, driven by the rapid convalescence of agents with strong and normal immunity. After day 10, both β and γ decline swiftly toward zero. The more pronounced drop in β corresponds to a shift in symptomatology—particularly the emergence of pneumonia and prolonged cases in the weak-immunity subgroup—while the fall in γ reflects the diminishing pool of recovering individuals as infections wane.

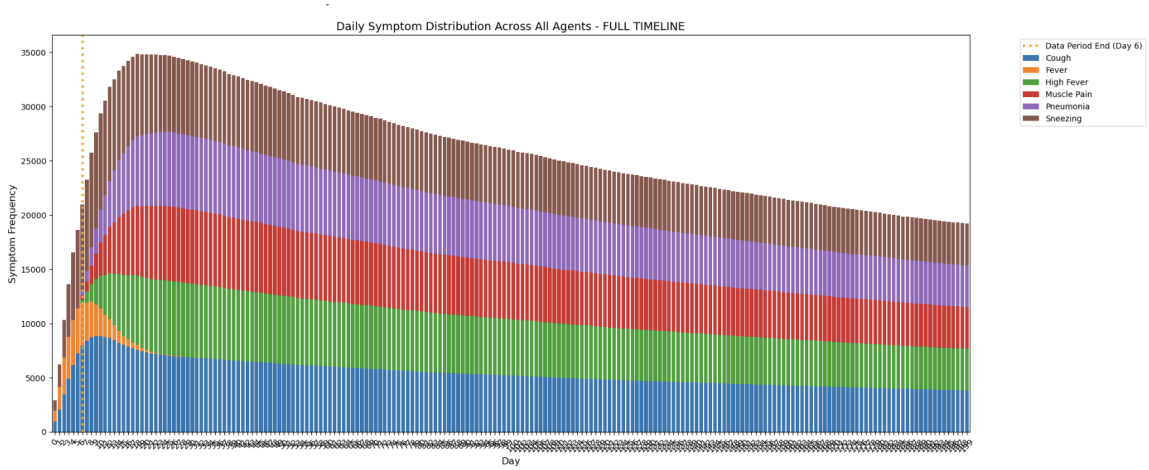


Figure 6: Daily Symptom Frequency.

This time-series plot depicts the progression of a lower respiratory tract infection in a population composed of 40% strong, 50% normal, and only 10% weak-immunity agents: during the first six days, a rapid increase in cough (blue) and fever (orange) cases drives the total symptom count up to 35 000 per day, with sneezing (brown) and muscle pain (red) also contributing in the early phase. However, because pneumonia (purple) and high fever (green) predominantly occur within the weak-immunity subgroup, their rise does not sustain transmission at high levels—indeed, inspection of the transmission and recovery rates (β/γ) shows that after day 10, β rapidly falls toward zero, indicating that new infections largely cease. Consequently, although the prolonged tail in symptom frequency reflects the extended course of severe cases among the small weak cohort, the depletion of susceptible contacts brings the outbreak to a close. This underscores that ultimate epidemic control is governed primarily by the broader population's immunity profile and the transmissibility potential associated with each symptom type.

Figure 7's Kaplan–Meier survival curves provide a comprehensive view of morbidity and mortality over time in lower respiratory tract infection: in the left panel, the overall survival probability declines steadily from 1.00 at infection onset to approximately 0.83 by day 200—reflecting the cumulative burden of complications and late-stage sequelae—and the narrow shaded confidence interval attests to the statistical robustness of the dataset; in the right panel, stratification by treatment setting reveals that patients managed at home (orange curve) maintain a consistently high survival rate ($\sim$ 0.91–0.92) throughout the follow-up period, whereas hospitalized severe cases (blue curve) experience a markedly steeper decline, reaching roughly 0.70 by day 200. A log-rank test ($p = 0.0000$) confirms that these intergroup survival differences are highly statistically significant. Collectively, these results highlight the necessity of nuanced clinical pathways in LRTI management: by prioritizing outpatient monitoring and supportive care for mild to moderate cases, healthcare systems can conserve critical inpatient resources, while deploying early intensive care interventions and advanced therapeutic modalities for

the most severe hospitalized patients, thereby optimizing both epidemic control and individual patient survival outcomes.

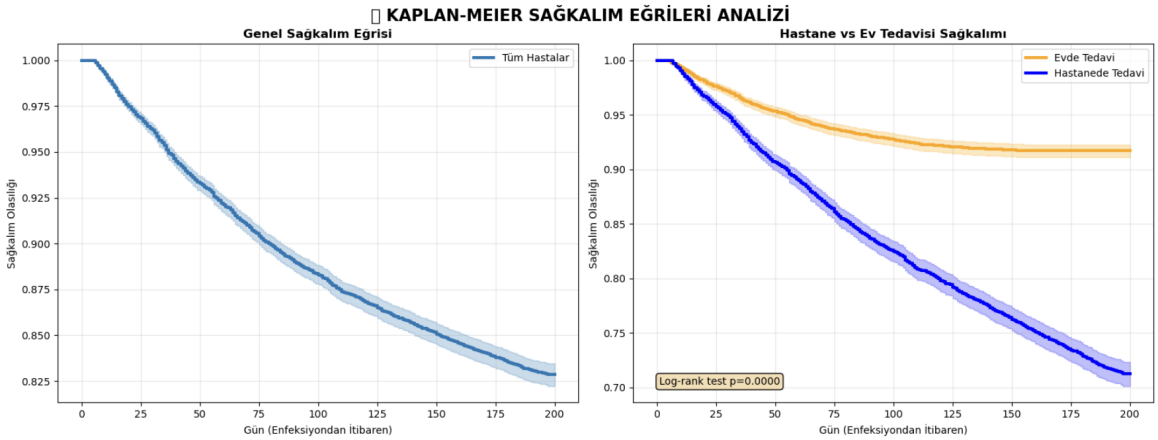


Figure 7: Kaplan-Meier's Curves.

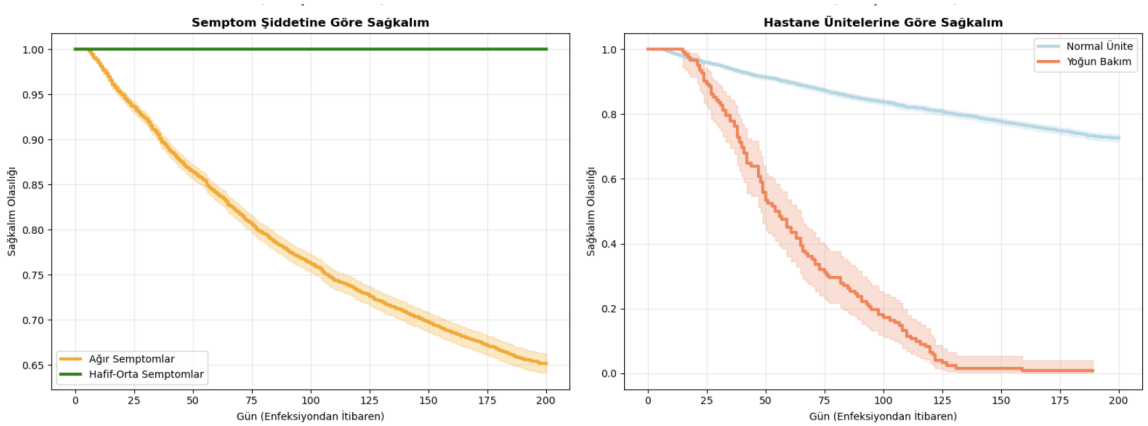


Figure 8: Kaplan-Meier's Curves.

Figure 8's Kaplan–Meier survival curves integrate symptom severity and hospital unit stratification into a unified perspective: patients with mild to moderate symptoms (green curve) maintain a 100% survival rate throughout the 200-day follow-up, indicating negligible mortality in this cohort, whereas the severe symptom group (orange curve) exhibits a steeply declining survival trajectory from the outset, reaching approximately 65% by day 200—underscoring the direct correlation between symptom severity and prognosis. In the right panel, general ward patients (light blue curve) demonstrate a gradual decline in survival to around 75% by day 200, while intensive care patients (dark orange curve) experience a dramatic drop, with survival rates falling to 20% within the first 100 days and approaching 0% by day 150, highlighting the elevated mortality risk among critical cases. These findings emphasize the vital importance of triage strategies based on symptom severity and care setting: managing mild to moderate cases on an outpatient or general ward basis can preserve critical care resources for the most severe patients, thereby optimizing both epidemic control and patient survival outcomes.

This violin plot illustrates the distribution of recovery times (in days) for the three immunity groups:

Strong Immunity (green): The median recovery time is around 6 days, with the bulk of cases falling between 3 and 10 days. The distribution is narrow, indicating consistently rapid convalescence, although rare outliers extend the tail up to 200 days. Normal Immunity (blue): Displays a similar profile to the strong group, with a median near 6–7 days and most recoveries occurring within 3–10 days. While the tails also reach up to 200 days, the density is concentrated around the same short

recovery window. Weak Immunity (red): Shows a higher median of about 8 days and a much wider distribution. The violin's body spans roughly 5–20 days, but the long upper tail indicates meaningful density even up to 100 days or more, reflecting substantial variability and the potential for prolonged, chronic courses in this subgroup.

While the strong and normal immunity cohorts demonstrate predictable and brief recovery periods, the weak immunity subgroup not only has a longer median convalescence but also exhibits significant heterogeneity and long-tailed delays, underscoring how a small vulnerable segment can substantially extend both individual recovery and overall epidemic duration.

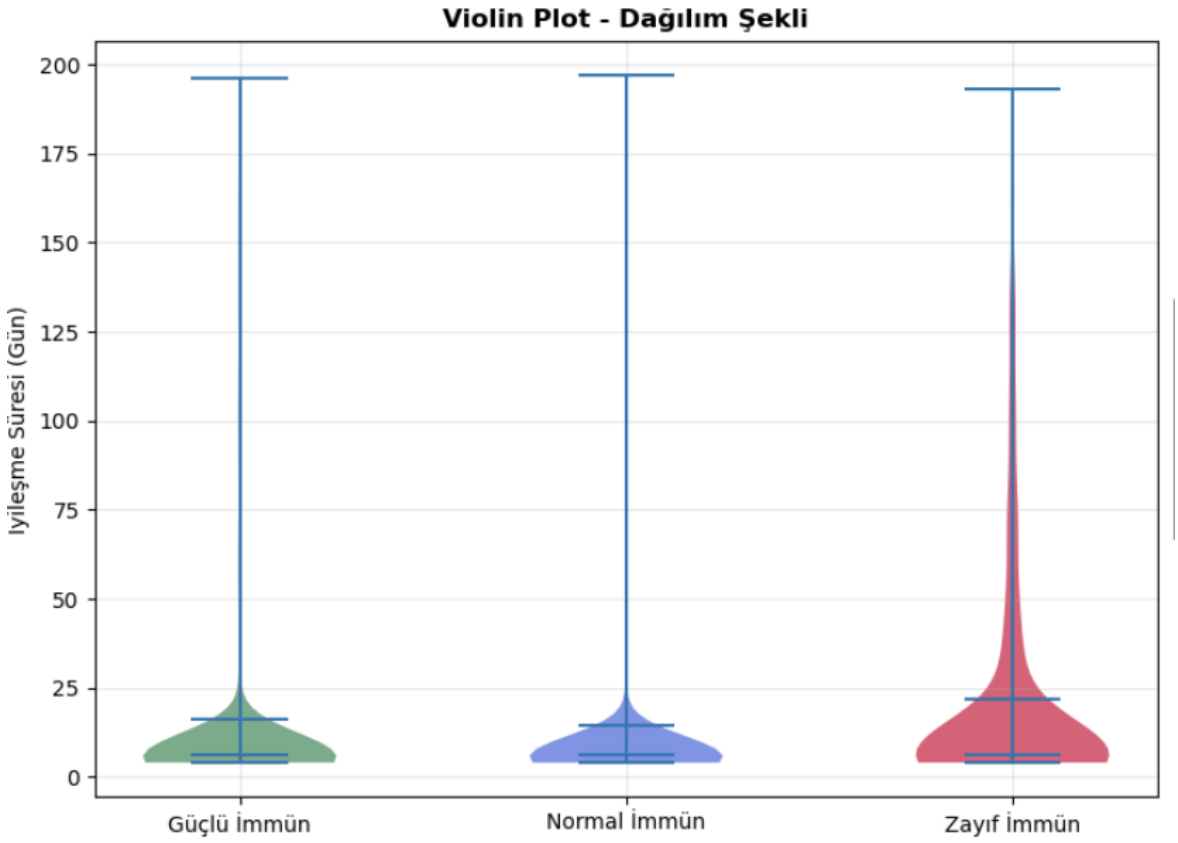


Figure 9: Recovery Rates.

3.3 MeV-1 Like Infection

The film *Contagion* (2011) is a realistic pandemic thriller that depicts a worldwide outbreak caused by a fictional virus called MeV-1, which has an unusually high mortality rate. One of the main criticisms leveled at the film is that it would be highly unlikely for a disease with such a high fatality rate to spark a global pandemic in the absence of any obvious transmission vector. In our simulation, we investigated how a MeV-1-like pathogen might spread and arrived at the following results:

Table 4: Symptom-based transmission parameters for the severe LRTI simulation (first 5 days)

Day	**Symptoms**	β_{combined}	β_{SIR}	ζ_{raw}	ζ_{net}	γ_{SIR}
0	Cough, Fever, Runny Nose	1.5696	0.6108	1.0920	0.7904	0.4000
1	Cough, Fever, Runny Nose	1.5696	0.6108	1.0920	0.7904	0.4000
2	High Fever, Muscle Pain	0.0000	0.0048	0.8060	0.7518	0.1967
3	Loss of Consciousness, Fever, Muscle Pain	0.0000	0.0046	1.9314	1.7242	0.0050
4	Coma, Sepsis	0.0000	0.0017	1.9670	1.9283	0.0022

Based on Figure 10, we can say that the disease infection period lasts approximately seven days, with no transmission occurring outside that window. The mortality rate is, as expected, extremely high and severe symptoms greatly inhibit recovery, causing the vast majority of infected agents to die.

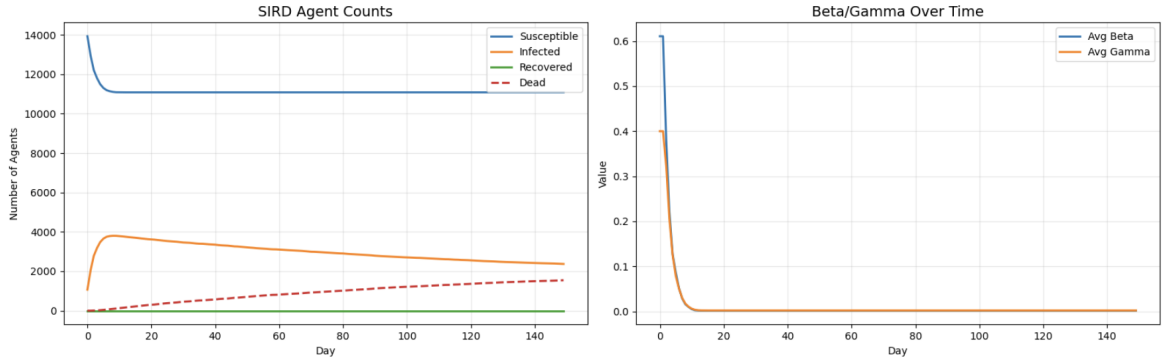


Figure 10: SIRD Graph and Average β and γ Values .

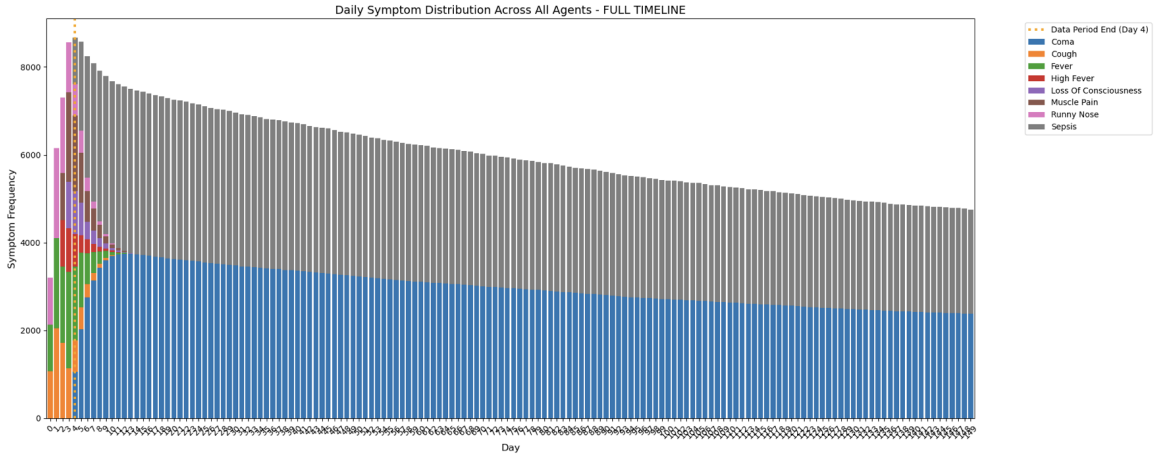


Figure 11: Daily Symptom Frequency.

In our model, the MeV-1–like virus is defined with such a high fatality rate that infected individuals develop severe symptoms and rapidly progress to a critical state, leaving them neither the time nor the opportunity to transmit the virus to others. Unlike a typical outbreak, this breaks the chain of person-to-person transmission: as soon as a host enters the heavy-symptom phase, they are isolated or require urgent care, and virtually no asymptomatic or mildly symptomatic carriers emerge in the population. Consequently, the vast majority of those infected follow a clinical course that ends directly in death, effectively severing the virus's transmission chain. Rather than spreading broadly through the community, the epidemic self-limits due to the abrupt fatalities at its onset. This scenario starkly illustrates the inherent contradiction of sustaining a pandemic that is simultaneously highly transmissible and highly lethal.

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